

Data Structures and Algorithms for Problem Solving (CS1.304)

Slip Test (17 Aug 2026)

Instructor: Kshitij Gajjar

Similarity between Words

Given two words **A** and **B**, their (Levenshtein) distance is the minimum number of edits needed to transform **A** to **B**. Edits are of three types:

1. Letter insertions (denoted by ✓)
2. Letter deletions (denoted by ✗)
3. Letter substitutions / replacements (denoted by ♻️)

Notation for distance:

$\text{dist}(\mathbf{A}, \mathbf{B})$.

Example 1

$\text{dist}(\text{croak}, \text{roast}) = 3.$

c	r	o	a	k	✓
✗	r	o	a	s	t

croak

↓ ✗_c

roak

↓ k♻️s

roas

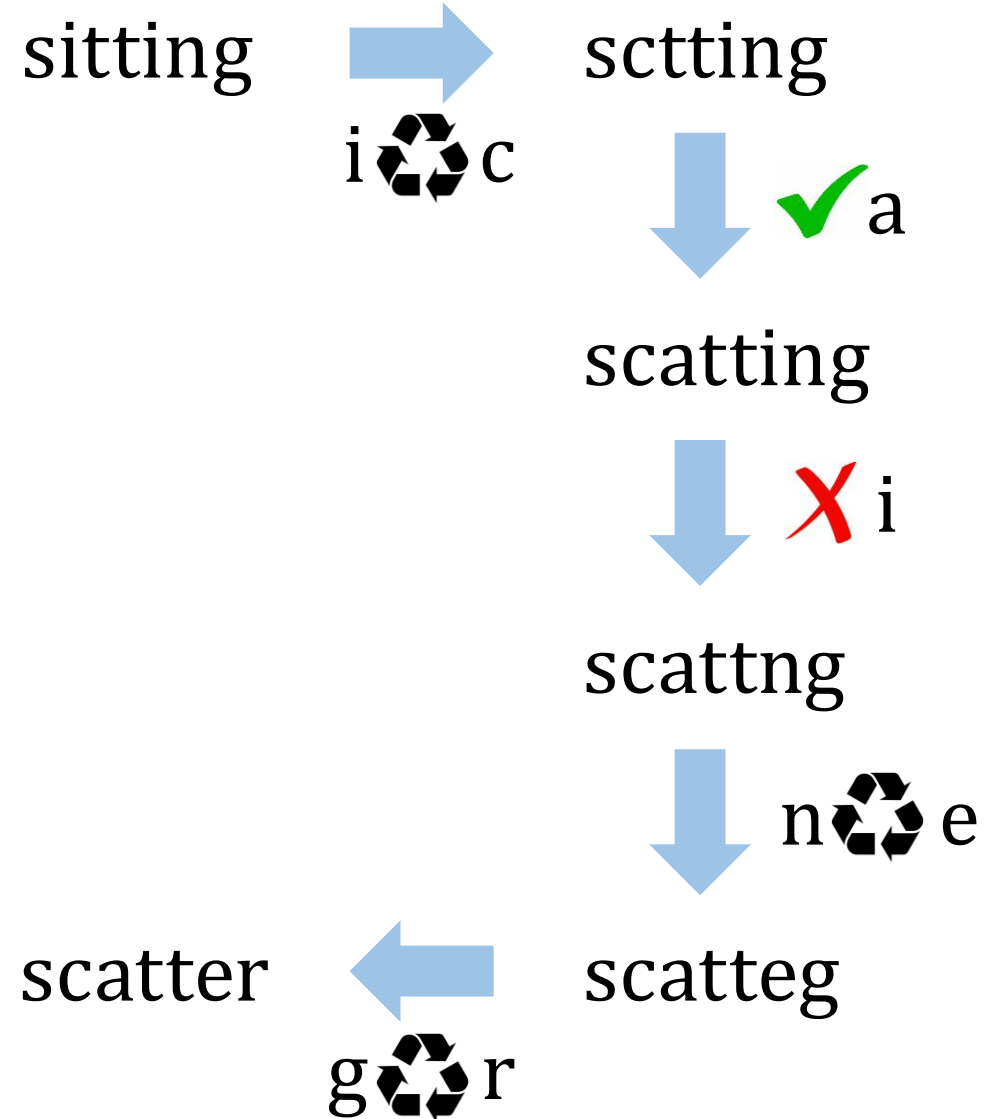
↓ ✓_t

roast

Example 2

$\text{dist}(\text{sitting}, \text{scatter}) = 5.$

s	i	✓	t	t	i	n	g
							
s	c	a	t	t	✗	e	r



Puzzle 1 [3 marks]

Evaluate the following. Write down your answers using the notation explained earlier.

- `dist(sleeveless, lessens)`
- `dist(intelligent, belligerent)`
- `dist(melanin, millennial)`

Puzzle 2 [7 marks]

Devise an algorithm to find the distance between two strings ***A*** and ***B*** of lengths ***n*₁** and ***n*₂**, respectively.

You shall be graded on the following parameters:

- Your code / pseudocode.
- Your explanation of its working.
- Your analysis of its runtime.

Thank You!

